
chapter six

CONTINUOUS RANDOM VARIABLES AND THE NORMAL DISTRIBUTION

*Everybody believes in the [normal approximation],
the experimenters because
they think it is a mathematical theorem,
the mathematicians because
they think it is an experimental fact.*

– G. Lippmann

INTRODUCTION

Many of the ideas in this chapter on **continuous probability distributions** are extensions of material we have already seen in the chapter on discrete probability distributions. The idea of probability as area will again be used. But the area will be under a curve rather than in the rectangles of the probability histograms we used in the last chapter.

There are several key differences in this chapter. In very simple terms, a discrete variable meant that we were counting something. In this chapter, a **continuous variable** is the result of measuring something rather than counting. There is a major difference in the probability statements we will make using discrete and continuous variables.

In a discrete probability distribution, it was possible to find the probability of getting a specific value of the random variable. When we

work with a continuous random variable, the probability of the variable being equal to a specific value is zero. But we are able to find the probability that a variable is in an interval. The **Empirical Rule** will be used to find the probability that the continuous variable is in an interval about the mean.

This chapter provides an important transition between the idea of hypothesis testing and a probability distribution; first introduced in the last chapter and then extended to continuous variables in this chapter. Throughout this chapter, we will use the z score that gives the distance between the mean and the point represented by z in terms of the standard deviation. We will refer to z as the **standard score**; it is also known as the standard normal variable. The z score and the bell shaped **standard normal distribution** become the focal points in this chapter. The z score will be used to find the area under the **standard normal curve** that is used to find probabilities

The z score will continue to be a focus in the next several chapters. We will use the z score, in various forms, as a test statistic to do hypothesis testing in inferential statistics. The common structure of the various forms of the z score as a test statistic will help us organize the material in the chapters that follow.

In this chapter, notation continues to be very important to us as a way of organizing our work and understanding what it is that we have done. We will also do a great deal of drawing; it is important that we draw a picture of a bell-shaped curve for each problem that we do.

The last thing we will look at in this chapter is the **normal approximation** to the binomial distribution. Recent advances in calculators and the use of statistical software on personal computers make this section less important than it would have been in the past decade. But the approximation will help us better understand the relationship between the discrete and continuous random variable.

CHAPTER 6 HOT SPOTS

1. **A Difference Between The Probability For Discrete And Continuous Random Variables.**
Starts on page 6–4. Mann 6.1
2. **Understanding Why The Probability Of A Continuous Random Variable Equalling A Specific Value is Zero.**
Starts on page 6–5. Mann 6.1
3. **Using The Empirical Rule To Find $P(a \leq x \leq b)$.**
Starts on page 6–5. Problems on pages 6–6, 6–7, 6–8. Mann 6.3
4. **Using The Z Score To Find $P(x)$.**
Starts on page 6–8. Problems on pages 6–10, 6–11. Mann 6.4
5. **Using Graphs With The Z Score.**
Starts on page 6–11. Problems on pages 6–14, 6–15.
Mann 6.3, 6.4, 6.5
6. **Finding c For $P(z \geq c) = 5\%$.**
Starts on page 6–15. Problems on pages 6–17, 6–18. Mann 6.6
7. **Normal Approximation To The Binomial Distribution.**
Starts on page 6–18. Problems on pages 6–21, 6–22. Mann 6.7

If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic for a help session with your professor.

Hot Spot #1 – A Difference Between The Probability For Discrete And Continuous Random Variables

In this chapter we work with many of the same ideas we developed and used in the chapter on discrete probability distributions. However, there is one very important difference beyond the over simplified distinction that we use a discrete random variable when we count and a continuous random variable when we measure. In the work we did with discrete probability distributions, we were able to find the probability that a random variable equals a specific value. But, when we find the probability of a continuous random variable having a specific value, we get zero. (This is discussed further in **Hot Spot #2**).

For example, if the random variable x has a binomial distribution with $n = 5$ and $p = 0.5$, we can find $P(x = 3)$. We find the answer by substituting the values of n and p into the binomial probability function

$$P(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

In this example, we get

$$P(x = 3) = 10 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = 0.3125$$

$$\text{where } 10 = \frac{5!}{3!2!}$$

If x is a continuous random variable, the probability that x equals a specific value is zero. Using notation, we would write $P(x = c) = 0$, where c is any real number.

When we work with a continuous random variable, we are only able to find the probability that x is in an interval. For example, we are able to ask for the probability that x is between a and b . The notation we use involves the expression $P(a \leq x \leq b)$.

We then find the area between a and b under the curve defined by the continuous probability distribution.

Hot Spot #2 – Understanding Why The Probability Of A Continuous Random Variable Equalling A Specific Value Is Zero.



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Section 6.1

It is sometimes difficult to understand why the probability of a specific value is zero. One way of looking at the problem is to consider a large paper bag in which we write the ten digits between 0 and 9 on slips of paper and place them in the bag. If we then draw out one value, the probability of getting one of the digits between 0 and 9 would be 0.1. Now if we consider adding slips of paper to represent the “tenths” place we could find the probability of getting a specific value to the tenths place between 0 and 9.9. For example we could find the probability of getting a value like 3.4, but now we would have one hundred slips of paper in the bag. The probability of getting one of the values in the bag is now 0.01.

We could then add more slips of paper to the bag for the “hundredths” place so that we can find the probability of getting a value like 3.44. But the probability of getting a specific value is now down to 0.001. As we continue adding slips of paper to the bag to allow us to get the probability of getting any real number between 0 and 10, we would get smaller and smaller probabilities that would get closer and closer to zero. As you realize, we would never be done adding slips of paper to our bag; and we would finally simply say that the probability of getting a specific real number in the half-open interval of 0 to 10 is zero.

Hot Spot #3 – Using The Empirical Rule to Find $P(a \leq x \leq b)$



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Section 6.3

We are usually interested in finding the probability that a continuous random variable is within some interval. For example, let x equal the speed of a car on a highway in miles per hour. If x is assumed to be normally distributed with $\mu = 55$ and $\sigma = 5$, then we could find the

probability that the speed of the car is more than 65.

There are two ways for us to look at this problem: using the Empirical Rule, or using a z score; which is the topic of the next **Hot Spot**. In this **Hot Spot**, we will use the Empirical Rule. Although this will be easy to do, we must realize our answer will only be an approximation of the correct answer we obtain using the z score.

We know that 95% of the speeds will be within two standard deviations of the mean. The mean is 55 mph and two standard deviations gives us 10 mph. When we use 55 for the mean, and then add and subtract the 10, we get the values of 45 and 65. So, 95% of the speeds are in the interval (45,65).

This leaves approximately 5% of the area under the bell-shaped curve outside the interval from 45 to 65. The areas outside of the interval are called the tails of the curve. The area under the right tail of the curve gives the probability of $x \geq 65$. The area under the right tail of the curve is one-half of 5%, and we can write:

$$P(x \geq 65) = 2.5\%$$

Hot Spot #3 Sample Problem and Solution: Use the Empirical Rule to find the approximate answer for $P(50 \leq x \leq 60)$ where the continuous random variable x equals the speed on the highway with $\mu = 55$ and $\sigma = 5$.

Solution: The interval between 50 and 60 is centered on 55. The graph for the bell-shaped curve centered on 55 follows (Figure 6-1):

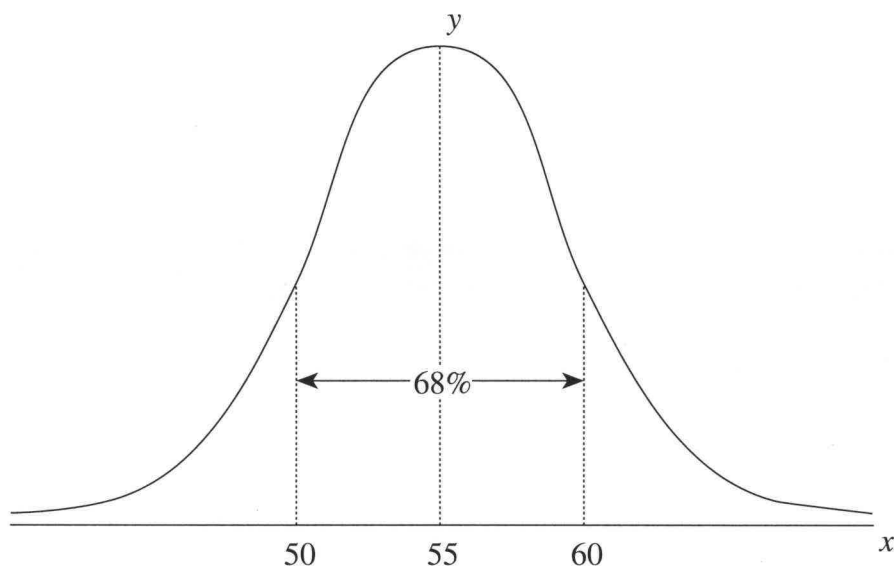


Figure 6-1:
One Standard
Deviation From
Mean of 55

The values of 50 and 60 represent the mean plus and minus one standard deviation. We know that approximately 68% of the distribution will be within one standard deviation of the mean. We can now answer the question using the appropriate notation:

$$P(50 \leq x \leq 60) = 0.68$$

Sample Problem and Answer: Let x be a continuous random variable with a normal distribution with $\mu = 1,200$ and $\sigma = 100$. Use the Empirical Rule to find an approximate value for $P(x \geq 1,300)$.

Answer: $P(x \geq 1,300) = 0.16$.

Sample Problem and Answer: Let x represent the scores on a statistics test given to all the statistics students at a university. Assume that x has a normal distribution with $\mu = 70$ and $\sigma = 10$. Use the Empirical Rule to find the approximate probability for $P(60 \leq x \leq 90)$.

Answer: $P(60 \leq x \leq 90) = 0.34 + 0.475 = 0.815$.

Hot Spot #4 – Using The Z Score To Find $P(x)$

An alternative to using the Empirical Rule to find $P(x)$ is to trade in questions about x for questions about z . The z score compares a data value's relationship to the mean in terms of the standard deviation. We do this so that we can obtain more accurate answers and make it easier to get the answers. Rather than having to figure out in our head how many standard deviations 65 mph is from 55 mph in our example about highway speeds with $\mu = 55$ and $\sigma = 5$, we just use the formula

$$z = \frac{x - \mu}{\sigma}$$

We substitute 65 for x and use the values of μ and σ given:

$$z = \frac{65 - 55}{5} = 2$$

The $z = 2$ tells us that we are two standard deviations away from the mean. Now we can replace our original question about the

$$P(x \geq 65) \text{ with } P(z \geq 2)$$

Remember, it does not matter if we use $\geq$ or just $>$ because the

$$P(x = 65) = 0 \text{ and } P(z = 2) = 0$$

Furthermore, it does not matter whether we use the x value or the z value. Either way, we want the area under the appropriate bell-shaped curve as shown in Figure 6-2:

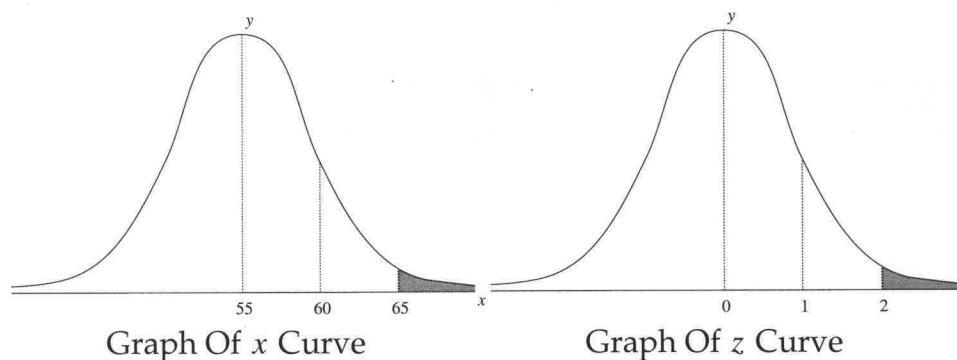


Figure 6-2:
Graphs Of x And
 z Distributions
Compared

The real problem is to find the area under the x curve to the right of 65. There are several methods that can be used, and some methods would take a great deal of time or require a computer to get an accurate answer. For example, one of the methods we could use involves a numerical technique of integration that is done in calculus.

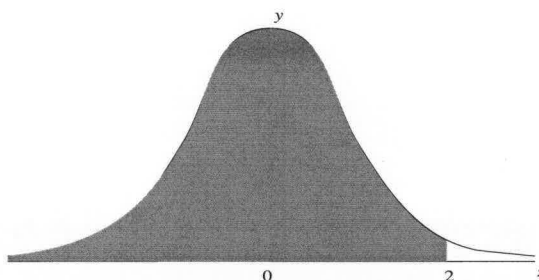
Fortunately, the z curve gives us an answer to the problem very quickly. All we have to do is use the z table inside the front cover of the text. We look up the value of 2 and find that the probability of z being to the left of 2 is 0.9772 (to the nearest ten thousandth).

To find the area under the curve to the right of $z = 2$, we subtract our value of 0.9772 from 1 as follows:

$$P(z \geq 2) = 1 - 0.9772 = 0.0228$$

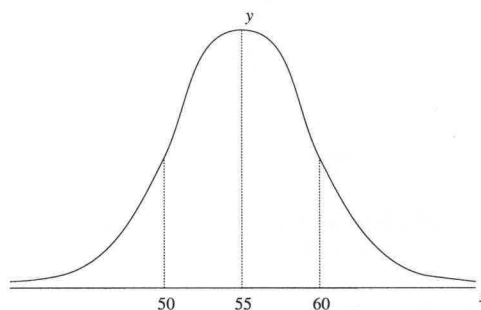
Figure 6-3, below, shows the bell-shaped curve for z and the area under the curve to the left of 2. As is illustrated, we subtract the area that is shaded from 1 to get the area to the right of 2.

Figure 6-3: Area Under The Curve to the left of 2



Compare the next three problems to the corresponding problems of **Hot Spot #3** on pages 6-6, 6-7, and 6-8.

Figure 6-4: Bell Curve Centered At 55



Hot Spot #4 Sample Problem and Solution: Use the z value to find the exact answer for $P(50 \leq x \leq 60)$; where the continuous random variable x equals the speed on the highway with $\mu = 55$ mph and $\sigma = 5$ mph.

Solution: Step 1. Graph the bell-shaped curve centered on 55 mph.

Step 2. Place the values of 50 and 60 on the graph.

Step 3. We want the area under the curve between the two values. We now find the z values for each value of x .

$$z = \frac{50 - 55}{5} = -1 \qquad z = \frac{60 - 55}{5} = 1$$

Step 4. We now use the table of z values to find the area between -1 and 1. We want to find the following:

$$P(-1 \leq z \leq 1) = P(z \leq 1) - P(z \leq -1)$$

Step 5. We subtract the area under the curve to the left of -1 from the area under the curve to the left of 1. We find these two areas in the z table and subtract.

$$\begin{aligned}P(50 \leq x \leq 60) &= P(z \leq 1) - P(z \leq -1) \\&= 0.8413 - 0.1587 = 0.6826\end{aligned}$$

Sample Problem and Answer: Let x be a continuous random variable with a normal distribution with $\mu = 1,200$ and $\sigma = 100$. Use the z score to find the exact answer for $P(x \geq 1,300)$.

Answer: $P(x \geq 1,300) = P(z \geq 1) = 1 - P(z \leq 1) = 1 - 0.8413 = 0.1587$

Sample Problem and Answer: Let x represent the scores on a statistics test given to all the statistics students at a university. Assume that x has a normal distribution with $\mu = 70$ and $\sigma = 10$. Use the z score to find the exact probability for $P(60 \leq x \leq 90)$.

Answer: $P(60 \leq x \leq 90) = 0.9772 - 0.1587 = 0.8185$

Hot Spot #5 – Using Graphs With The Z Score

A very important point needs to be made about working with z values. **Always take time to draw a graph!** The picture we draw helps us see the areas we need to use to get an answer. For example, if we use the normal distribution for x , where x represents highway speeds with $\mu=55$ and $\sigma = 10$, we can find the probability that x is between 60 and 70.



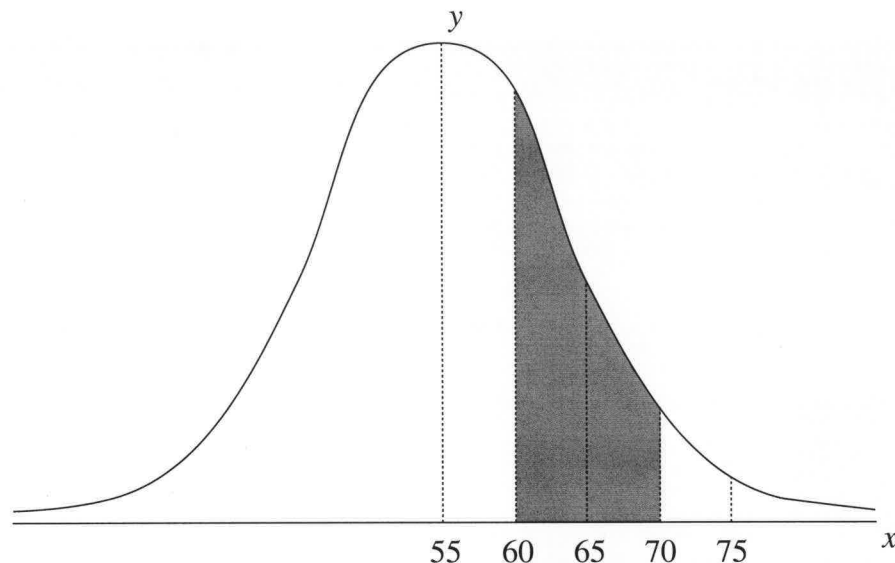
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Sections 6.3, 6.4,
And 6.5*

Figure 6-5:
Probability Of x
Between
60 and 70

We would write this using the following notation:

$$P(60 \leq x \leq 70)$$

The first thing we want to do is draw a diagram or graph:



Since 60 and 70 are both larger than the population mean of 55, the 60 and 70 are placed on the right side of the curve as shown. We want the area under the curve between 60 and 70. Using the type of approach we followed earlier, we could try to use the Empirical Rule. But this time, we find that we need the area under the bell-shaped curve between one-half and one and one-half standard deviations; $60 = 55 + 0.5(10)$ and $70 = 55 + 1.5(10)$. The Empirical Rule does not help in this situation. Unless we want to use a technique such as numerical integration from calculus, we have no choice but to use the z values that correspond to 60 and 70 as follows:

$$z = \frac{60 - 55}{10} = \frac{1}{2} = 0.5 \qquad z = \frac{70 - 55}{10} = \frac{3}{2} = 1.5$$

The graph we drew now helps us decide how to find the answer to our original question. Our question is now written as

$$P(0.5 \leq z \leq 1.5)$$

Using the z table inside the front cover of the text, we will get two different areas as shown below in Figure 6-6:

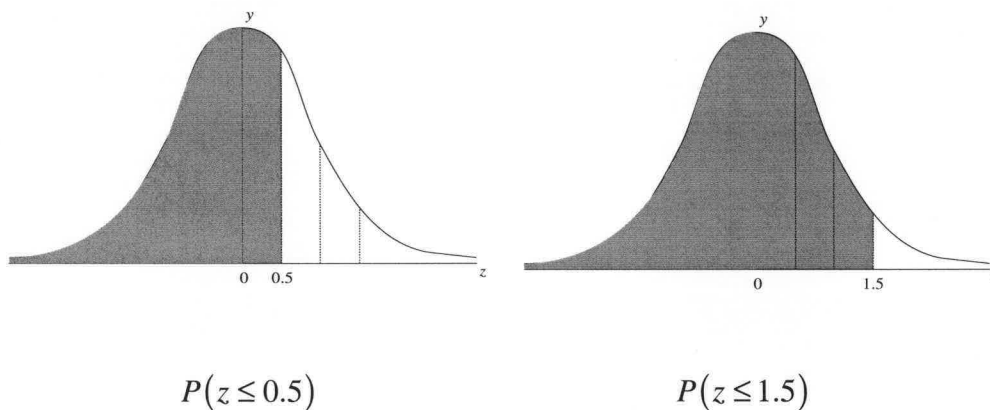


Figure 6-6:
Two Different z
Table Areas

We need to subtract the smaller area to the left of 0.5 from the larger area to the left of 1.5 to get our answer. We find the areas from the z table. Using the z table, we look up 1.5 and find the large area is 0.9332; it is $P(z \leq 1.5)$. When we look up 0.5 in the z table, we find the small area is 0.6915; it is $P(z \leq 0.5)$. We use the graph to show us that we need to find the difference of the two areas. Even if we somehow did the subtraction backwards, we would realize our mistake because **the probability must be a positive number**. We subtract as follows:

$$\begin{aligned}
 P(0.5 \leq z \leq 1.5) &= P(z \leq 1.5) - P(z \leq 0.5) \\
 &= 0.9332 - 0.6915 = 0.2417
 \end{aligned}$$

It is important to document how we get the answer using both the graph and probability notation. Many students try to take shortcuts at this point by either leaving out the notation or not drawing the graph.

Although it is possible to get the correct answer, students often get confused and make errors when they use shortcuts. So, be careful; document your work with a graph and probability notation.

Hot Spot #5 Sample Problem and Solution: Find $P(-2 \leq z \leq -0.5)$.

Solution: Step 1. Draw the bell-shaped curve.

Step 2. Locate -2 and -0.5 on the curve and shade in the area under the curve between the values.

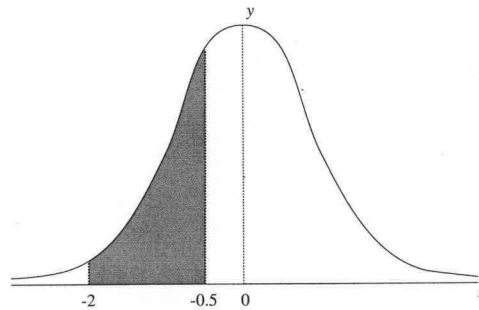


Figure 6-7:
Area Between
-2 and -0.5

Step 3. We see that we need to subtract two areas:

$$P(z \leq -0.5) - P(z \leq -2)$$

Step 4. Using the z table we subtract the following areas:

$$P(z \leq -0.5) - P(z \leq -2) = 0.3085 - 0.0228 = 0.2857$$

Step 5. We should rewrite our answer as follows:

$$\begin{aligned} P(-2 \leq z \leq -0.5) &= P(z \leq -0.5) - P(z \leq -2) \\ &= 0.3085 - 0.0228 \\ &= 0.2857 \end{aligned}$$

Sample Problem and Answer: Find $P(-1.5 \leq z \leq -0.5)$.

$$\begin{aligned}\text{Answer: } P(-1.5 \leq z \leq -0.5) &= P(z \leq -0.5) - P(z \leq -1.5) \\ &= 0.3085 - 0.0668 = 0.2417 \\ P(-1.5 \leq z \leq -0.5) &= 0.2417\end{aligned}$$

Sample Problem and Answer: Find $P(z \geq -1.5)$.

$$\begin{aligned}\text{Answer: } P(z \geq -1.5) &= 1 - P(z \leq -1.5) \\ &= 1 - 0.0668 = 0.9332 \\ P(z \geq -1.5) &= 0.9332\end{aligned}$$

Hot Spot #6 – Finding c For $P(z \geq c) = 5\%$

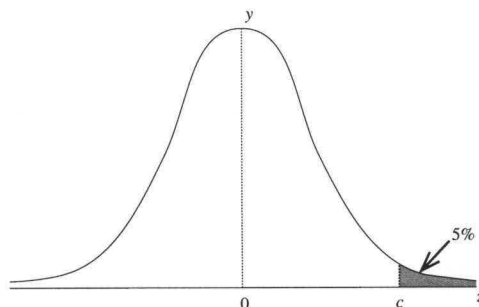
Related to the problem of finding the probability of z being between two values is the inverse problem of finding the value of z that gives a specific area to the right of z . For example, we may want to find the value of z that gives 5% to the right of our value. This problem can be stated in two different ways. The easiest way is to use the notation $z(0.05)$. There is a more interesting way of asking the questions as follows:

$$\text{Find } c \text{ so that } P(z \geq c) = 0.05$$

The letter c is used to represent the actual value of z that makes the statement true. In our example we know that c must be on the right

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Section 6.6

Figure 6-8: Use
Of c With z
Scores



side of the curve because the area we want to the right of c is less than 50%. If we now draw the graph with c on the right side, we get the following:

As is usually the case, it is very important to first draw the graph and locate c on the correct side. It is now easier to see that we need a value of c that gives 0.95 to the left of our value of c . We can use the z table to find that $c = 1.645$ gives us the 0.95 we need. We now replace the letter c with the 1.645 and write the following:

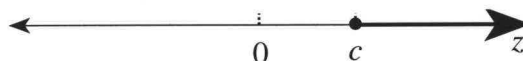
$$P(z \geq 1.645) = 0.05$$

Of all the concepts in this chapter on continuous probability distributions, this one idea seems to be very difficult to understand. Some of the difficulty seems to be with the algebra of the statement

$$z \geq c.$$

If we consider the above expression to be a problem with inequalities much like what we did in an algebra class, it may make it easier to understand what it means to have $z \geq c$. In the algebra class, we would just graph the inequality on a number line and the solution includes all of the values to the right of c as follows:

Figure 6-9:
 $z \geq c$ As An
Inequality On A
Number Line



Hot Spot #6 Sample Problem and Solution: Find the value of c that makes the following statement true: $P(-c \leq z \leq c) = 0.95$.

Solution: Step 1. We need to locate the values of c and $-c$ on the bell-shaped curve. It is easier this time because we have an interval. The graph is drawn below in Figure 6-10:

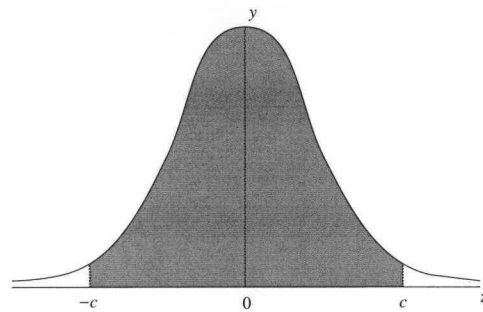


Figure 6-10:
Area Between
 $-c$ and c

Step 2. Since our values of c and $-c$ are symmetric to the center of the curve, we see that we need a value of c that will give us 0.975 to the left of our value c . We now look up 0.975 as an area in the z table. We find a value of $c = 1.96$ gives us the required area.

Step 3. We can now write the following:

$$P(-1.96 \leq z \leq 1.96) = 0.95$$

Step 4. Our solution is $c = 1.96$.

Sample Problem and Answer: Find the value of c that makes the following statement true: $P(z \leq c) = 0.01$.

Answer: $c = -2.326$



Sample Problem and Answer: Find the z value for $z(0.10)$.

Answer: $c = 1.282$

Hot Spot #7 – Normal Approximation To The Binomial Distribution

The last consideration in this chapter on continuous probability distributions involves what is called the normal approximation to the binomial distribution. The normal approximation is very useful if we do not have either a calculator or a computer that is programmed with the binomial distribution. When we have to rely on a binomial table to find probabilities, this approximation is used when either the number of trials is larger than a value listed in a table of binomial probabilities or when the p value is a value not listed in the table. The usual difficulty is illustrated in the following example.

Suppose we are asked to find the probability of getting a score of 70 or more on a 100 question True/False test and that we must assume that the individual taking the test is guessing on every question. If we let x = the number correct, then x is a discrete random variable with $n = 100$ and $p = 0.5$; and we want the $P(x \geq 70)$.

The partial probability histogram below (Figure 6–11) shows the areas we want between 70 and 100; the area includes the rectangles at $x = 70$ and $x = 100$. Whenever np and nq are both greater than 5, the histogram will be approximately bell-shaped. In our example $np = nq = (100)(0.5) = 50$; so we can use the normal approximation. A bell-shaped curve has been drawn over the probability histogram. The area under the curve will be approximately equal to the area in the rectangles, and we will use the normal curve to approximate the areas in the probability histogram. The curve will be centered at $\mu = np$, the mean of the binomial distribution.

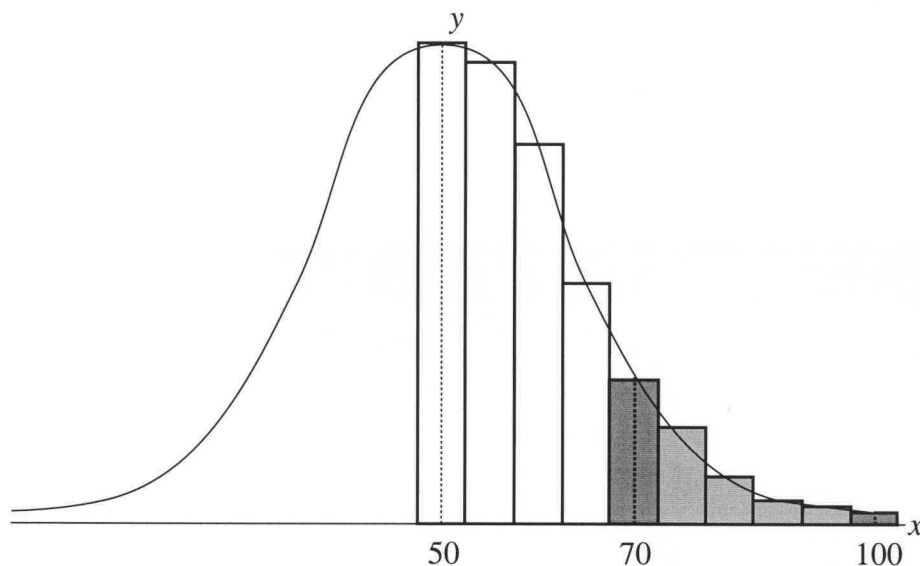


Figure 6-11:
Normal
Approximation

The area we want includes the rectangles at $x = 70$ and $x = 100$. The rectangle at 70 includes values on the x -axis from 69.5 to 70.5, and the rectangle at 100 includes the values on the x -axis from 99.5 to 100.5. If x is a continuous variable, we want the area under the curve to the right of 69.5 and to the left of 100.5. This is often a confusing point; the important thing to understand is that the following two statements are referencing the same areas:

If x is a discrete variable, we want $70 \leq x \leq 100$.

If x is a continuous variable, we want $69.5 \leq x \leq 100.5$.

Although it looks like we are talking about the same variable we are not. In $70 \leq x \leq 100$, the variable is discrete and it must be a positive integer. In $69.5 \leq x \leq 100.5$, the variable is continuous; there is no longer a restriction that it be an integer. To find the approximate answer to our problem using the normal curve we find

$$P(69.5 \leq x \leq 100.5) \text{ where } \mu = np = 50 \text{ and } \sigma = \sqrt{npq} = 5$$

We now must find the values of the related z scores:

$$z = \frac{69.5 - 50}{5} = 3.9 \qquad z = \frac{100.5 - 50}{5} = 10.1$$

We want the area under the normal curve between 3.9 and 10.1; however, the area to the right of $z = 10.1$ is so small that we can ignore it in this case. We will simply use the area to the right of 3.9; since we can not find 3.9 in the z table we can use the HP21 to find the area to the left of 3.9 and subtract the value from 1 to get our answer:

$$\begin{aligned} P(x \geq 70) \text{ for discrete variable} &= P(x \geq 69.5) \text{ for continuous variable} \\ &= 1 - 0.99995 = 0.00005 \end{aligned}$$

If we use the z table we find that 3.49 is the largest value of z in the table. The area to the left of 3.49 is 0.9998; so the area to the right of 3.49 gives us 0.0002 and all we can say is that the probability is less than 0.0002.

Clearly, if we have a computer program such as MINITAB we can get a correct answer without using the normal approximation. But the normal approximation to the binomial distribution provides us with a good opportunity to better understand both distributions.

One more point needs to be addressed regarding the normal approximation. If we had been asked to find the probability of getting more than 70 correct, we could again ignore the area to the right of 100 and write the expression

$$P(x_{\text{discrete}} > 70) = P(x_{\text{continuous}} \geq 70.5)$$

The rectangles for $x > 70$ include rectangles to the right of 70.5 because the first discrete value larger than 70 is 71. It includes the rectangle from 70.5 to 71.5.

Whenever we have a question as to what value to use in the approximation, it helps to draw the partial probability histogram and label it carefully to see which rectangles we need to use. As we can see

from the diagram below, it is clear that we need to use 70.5 for the x value of the continuous variable.

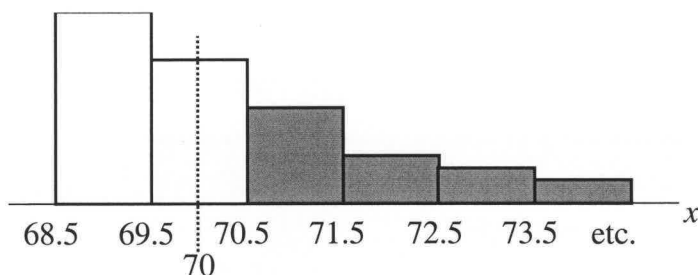


Figure 6-12:
Continuity
Correction
Factor

Hot Spot #7 Sample Problem and Solution: Given a 50 question three-part multiple choice test, what is the probability of getting less than 25 correct when guessing? (Include the area to the left of $x = 0$).

Solution: Step 1. $n = 50$ $p = \frac{1}{3}$ and $q = 1 - p = \frac{2}{3}$

Step 2. $np = \frac{50}{3} = 16.\bar{6}$ and $nq = \frac{100}{3} = 33.\bar{3}$

It is OK to use the normal approximation.

Step 3. $\mu = np \approx 16.7$ $\sigma = \sqrt{npq} = \sqrt{\frac{100}{9}} = \frac{10}{3} \approx 3.3$

Step 4. We want $P(x < 25) = P(x \leq 24)$

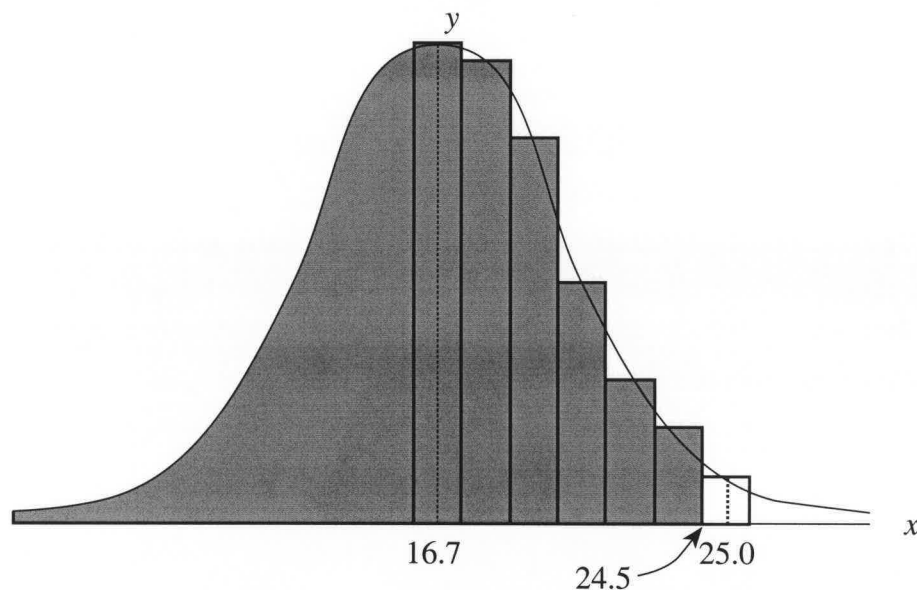
Step 5. Draw a partial graph of the histogram and curve to determine the value of the continuous variable x (See Figure 6-15).

Step 6. We want $P(x \leq 24.5)$ and we need to find the related value of $z = \frac{24.5 - 16.7}{3.3} = 2.36$.

Step 7. $P(z \leq 2.36) = 0.9909$

Step 8. $P(x < 25) = P(z \leq 2.36) = 0.9909$.

Figure 6-13:
Probability Of
Getting Less
Than 25 Correct
When Guessing



Sample Problem and Answer: Given a binomial random variable with 60 trials and $p = \frac{1}{6}$ and $q = 1 - p = \frac{5}{6}$, find the probability that x is greater than 20. (Include the area to the right of 20). Since the value of z we get is off the table, we will have to state our answer as an inequality.

Answer: $P(x > 20) = P(z \geq 3.64) \leq 1 - P(z \leq 3.49) = 1 - 0.9998 = 0.0002$

Sample Problem and Answer: Given a 40 question True/False test, find the probability of getting between 20 and 30 (inclusive) questions correct when guessing. Hint: Find the z value for $x = 19.5$ and $x = 30.5$.

Answer: $\mu = 40(0.5) = 20$ $\sigma = \sqrt{40(0.5)(0.5)} = 3.16228$

$z = \frac{19.5 - 20}{3.16228} = -0.16$ $z = \frac{30.5 - 20}{3.16228} = 3.32$

$P(-0.16 \leq z \leq 3.32) = P(z \leq 3.32) - P(z \leq -0.16)$
 $= 0.9995 - 0.4364 = 0.5631$

CHAPTER 6 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the difference between a discrete and continuous random variable?
2. Why is the probability of a specific value of a continuous random variable equal to zero?
3. What is the standard normal distribution?
4. What is the z score for the mean?
5. What is the relationship of the Empirical Rule to a z score?
6. When finding c so that $P(z \geq c) = 0.05$, how do you decide which side of the bell-shaped curve to place the letter c ?
7. Why is it important to graph the curve and locate the values on the curve when finding $P(1.5 \leq z \leq 2.5)$?
8. Given the continuous random variable x with $\mu = 55$ and $\sigma = 10$ why do we need a z score when finding $P(x \geq 57.5)$?
9. What is meant by the phrase "a family of normal distribution curves?"
10. When using the normal approximation to the binomial distribution, what is the correction for continuity?